

Technical Briefing: Unified Web3 & AI Ecosystem

Anant Aasha Foundation (FY 2026-2027)

1. Architectural Summary: Serverless Decentralized Stack

The Foundation is moving from a traditional hosting model to a governance-first, AI-native infrastructure built on the **Internet Computer Protocol (ICP)** and orchestrated by **n8n**. This stack ensures radical transparency for donors and operational efficiency for rural outreach without centralized points of failure.

- **Logic & App Builder: Caffeine.ai** serves as the primary "Self-Writing" interface to generate Motoko-based smart contracts (canisters) and the frontend dashboards.
- **Operational Control: n8n** acts as the "Legality Brain," governing process sequencing, third-party API interactions, and **Human-in-the-loop (HITL)** approvals.
- **Economic Model: Reverse Gas Architecture.** All computational costs (Cycles) are managed and pre-funded by the Foundation. End-users (students/parents) interact with the platform for free without needing to manage wallets or pay transaction fees.
- **Transparency Layer: On-Chain Impact Ledger.** Non-financial activity events (Seva proofs, course completions) are logged immutably on ICP for real-time donor reporting.

2. Core Functional Pillars

2.1 Three-Tier Multi-Tenant Dashboard

Development of three interlinked roles via Internet Identity (II) authentication:

1. **Admin/Teacher:** OCR upload of textbook chapters, student roster management, and manual approval of utility reward credits.
2. **Student/Parent:** Personalized learning paths using AI-generated visual materials and monitoring of "Aasha Coin" utility rewards.
3. **Donor:** Read-only portal providing "Impact ID" tracking and interactive maps of hardware deployment.

2.2 Content Factory & RAG (Retrieval-Augmented Generation)

An automated pipeline for converting physical textbooks into interactive EdTech assets:

- **Textbook Mandate:** For all subjects except Hindi and English Literature, modules must derive a **minimum of 60% of their content** directly from the uploaded source.
- **OCR to H5P:** Extraction of text/exercises via Google Vision API → Transformation into bilingual (Hinglish) visual notebooks (Flashcards, Infographics, Quizzes) and H5P interactive packages.

2.3 Multi-Tenant Smart Cache Logic

To scale across multiple institutions efficiently, the database utilizes a unique composite tagging system:

- **Composite ID:** School_ID + Class_ID + Subject_ID + Chapter_Number.
- **Reuse Logic:** New student registration triggers an automated lookup. If matching content exists in the "Global Cache," students are auto-linked to the pre-existing material to prevent duplicate AI compute costs and redundant OCR processing.

2.4 Gamified Assessment Engine

Transforming textbook exercises into high-engagement 2D action games (Tekken/WWE/Contra style):

- **Combat Mechanic:** Textbook-based questions are mapped to "Finishers" or special moves.
- **Verification:** Completion of missions triggers automated utility token rewards (Aasha Coins) upon human-in-the-loop verification via n8n.

3. Required IT Support Areas

We require expert support to implement and optimize the following components:

- **Motoko Canister Optimization:** Development of stable canisters for high-volume data storage (Smart Cache) and ensuring efficient cross-canister calls between the student and admin modules.
- **Complex n8n Workflow Design:** Building branching logic for:
 - **Tiered Logistics:** Integrating Shiprocket/Delhivery APIs for automated pickup scheduling (Bulk) vs. drop-off center location sharing (Small units).
 - **Compliance Guardrails:** Ensuring that utility coin distribution and statutory reporting data follow mandatory sequences.
- **Multimodal AI Logic:**
 - Implementing the **RAG grounding rules** (60% book source) within the content factory.
 - Configuring AI Vision within n8n workflows to label Jal-Seva proofs (e.g., detecting "birds" or "water bowls") before routing to human approval.
- **H5P Automation:** Programmatically generating valid .h5p packages (content.json and library structures) from AI-processed textbook text for embedding in the Caffeine-built dashboards.

4. Expert Discussion Points: Security & Trust

- **HITL (Human-in-the-loop):** Maintaining the principle that AI is assistive; final process authority resides in the n8n-governed human dashboard.
- **Data Isolation:** Enforcing tenant isolation so that student-parent data remains secure while utilizing shared global content caches.
- **Legacy-Web3 Fusion:** Utilizing Chain Fusion technology for future integrations (e.g., stablecoin-based sponsorships or cross-chain rewards).

Works cited

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